Journal of Perinatology www.nature.com/jp

REVIEW ARTICLE

Care of the critically ill neonate with hypoxemic respiratory failure and acute pulmonary hypertension: framework for practice based on consensus opinion of neonatal hemodynamics working group

Amish Jain [1], Regan E. Giesinger [2], Shyamala Dakshinamurti [3], Yasser ElSayed [3], Robert P. Jankov [4], Dany E. Weisz [1], Satyan Lakshminrusimha [5], Souvik Mitra [6], Mjaye L. Mazwi [1], Joseph Ting [7], Michael Narvey [3] and Patrick J. McNamara [2]✉

Circulatory transition after birth presents a critical period whereby the pulmonary vascular bed and right ventricle must adapt to rapidly changing loading conditions. Failure of postnatal transition may present as hypoxemic respiratory failure, with disordered pulmonary and systemic blood flow. In this review, we present the biological and clinical contributors to pathophysiology and present a management framework.

Journal of Perinatology (2022) 42:3–13; https://doi.org/10.1038/s41372-021-01296-z

The circulatory transition encompasses the critical period after birth when the pulmonary circulation must dilate to accommodate the entire cardiac output. Adaptation to extrauterine life requires a complex series of structural and functional changes in pulmonary vascular endothelium and smooth muscle, resulting in a precipitous decrease in pulmonary vascular resistance (PVR). Acute pulmonary hypertension of the newborn (aPH) is defined as a failure of the normal decline in PVR at or shortly after birth and may be associated with impaired right ventricular (RV) systolic performance [1]. We chose the term aPH, rather than persistent pulmonary hypertension of the newborn (PPHN) based on the fact that it takes approximately 14 days for PVR to fall in healthy babies after birth and more accurately reflects the clinical scenario of impaired oxygenation due to compromised pulmonary blood flow for the aforementioned reasons; hence, in essence, all infants have persistent elevation of pulmonary arterial pressure. In addition, persistence of the fetal shunts [patent ductus arteriosus (PDA); patent foramen ovale (PFO)] support systemic blood flow in patients with compromised RV systolic performance, albeit at the expense of oxygenation. This derailment results from the injury response of endothelial and smooth muscle cells to a multiplicity of perinatal stressors, including hypoxia, acidosis, inflammation, or direct lung injury such as pneumonia or meconium aspiration [2]. Failure of the normal decline in PVR after birth may be due to pulmonary vascular maldevelopment (e.g., closure of the ductus arteriosus in the fetus, chronic fetal hypoxia), neonatal maladaptation (e.g. meconium aspiration syndrome, perinatal asphyxia), or pulmonary underdevelopment (e.g., pulmonary hypoplasia or dysplasia secondary to oligohydramnios, congenital diaphragmatic hernia or a fetal lung mass). Knowledge of the biology of pulmonary vascular development enables scientific advancement and clinical innovation. The clinical consequences include hypoxemic respiratory failure and compromised systemic blood flow leading to impaired tissue oxygenation. Exclusion of congenital cardiac defects which mimic aPH and careful determination of the relative contribution of lung parenchymal disease are critical factors towards optimization of treatment approaches.

In the Spring of 2018, a group led by Drs McNamara and Jain assembled in Toronto (Canada), which included thought leaders in neonatal hemodynamics, to discuss the development of a standardized approach to the care of term infants with aPH. The following writing sub-groups were established to review the relevant literature and draft a relevant narrative: (i) Pulmonary vascular biology (RJ, SD); (ii) Pulmonary Vascular Physiology and heart function (MM, DW, SM, SL) ; (iii) Clinical diagnosis and monitoring (JT, MN); (iv) Role of Targeted Neonatal Echocardiography (TNE) evaluation (RG, YES); (v) Approach to treatment (RG, PM, AJ). The goal of the working group was to provide a comprehensive overview of state of knowledge and framework for clinical practice relevant to neonatologists who manage these critically ill patients.

[1]Department of Pediatrics, University of Toronto, Toronto, ON, Canada. [2]Department of Pediatrics, University of Iowa, Iowa City, Iowa, USA. [3]Department of Pediatrics, University of Manitoba, Winnipeg, MB, Canada. [4]Department of Pediatrics, University of Ottawa, Ottawa, ON, Canada. [5]Department of Pediatrics, University of California Davis, Sacramento, CA, USA. [6]Department of Pediatrics, Dalhousie University, Halifax, NS, Canada. [7]Department of Pediatrics, University of British Columbia, Vancouver, BC, Canada.
✉email: patrick-mcnamara@uiowa.edu

Received: 11 June 2021 Revised: 17 November 2021 Accepted: 2 December 2021
Published online: 11 January 2022

SECTION I: BIOLOGY OF PULMONARY VASCULAR DEVELOPMENT

- High fetal vascular tone persists throughout gestation through a variety of mechanisms. During the third trimester, muscularization of arteries occurs, enabling pulmonary vasoconstriction to be the primary determinant of high PVR. Neonatal pulmonary arterial smooth muscle exists in a low-energy tonic state to permit chronic vasoconstriction.
- At birth, lung expansion straightens kinked vessels, and reorganizes myocyte cytoskeletons into an elongated spindle shape. This combination enlarges the caliber of the vascular lumen. Simultaneously, reversal of hypoxemia and production of endogenous pulmonary vasodilators [e.g., nitric oxide, PGI_2, bradykinin] reduces vascular tone.

Anatomically, the mature pulmonary vasculature consists of elastic proximal pulmonary arteries and large muscular arteries, beyond which lie a vast network of small (<150 μm diameter) pre-capillary arteries, which are thin walled, partially- or non-muscular, have low basal tone, and are highly compliant. Mammalian lung development progresses through five overlapping phases: the embryonic, pseudoglandular, canalicular, saccular, and alveolar stages [3]. Pulmonary vessel formation accompanies budding and branching of the airways at every stage of lung development. It is during the canalicular phase that a primitive blood–gas interface is established that is sufficient to potentially support extrauterine life. The developing airways are often described as acting as a template for the development of the lung vasculature [4], due in part to the complex cross-talk that occurs between the branching airway and the surrounding mesenchyme. During the canalicular phase, the surrounding mesenchyme thins, bringing capillaries in closer apposition to the epithelium [5]. During the alveolar phase (commencing at approximately 32 weeks' gestation and proceeding well into the first decade of life), the mature blood–gas barrier emerges from coalescence of the double capillary network that arise within secondary septae to form a single capillary sheet [5]. High fetal PVR is maintained in the canalicular lung via low pulmonary circuit capacity. During the saccular stage, resistance falls as airway branching leads to vascular proliferation and thereby increased circuit capacity. Muscularization of arteries occurs during the alveolar phase of lung development, enabling the maintenance of high resistance through pulmonary vasoconstriction [6–9]. Neonatal pulmonary arterial smooth muscle differs from that of the adult in both its diminished contractile and relaxation capacity, suggesting the muscle exists in a low energy-utilizing tonic state [7], which facilitates the chronic maintenance of high vascular tone. Fetal pulmonary arteries also exhibit a brisk myogenic response to a pressure load, maintaining high resistance while opposing increased flow [10].

The triggers for pulmonary arterial relaxation, after birth, include tidal expansion of the lungs and reversal of relative hypoxemia, increased endogenous nitric oxide, PGI2 and bradykinin, and endocrine changes at parturition [11]. Some of the increase in circuit capacity is accounted for simply by the distension of the lungs with air, and the reduction in turbulence associated with straightening of kinked vessels [12]. Remodeling of the vascular wall also occurs after birth [13]. Fetal pulmonary arteries comprise brick-shaped smooth muscle cells arrayed in a tightly compressed media with a narrow arterial lumen. During transition, the myocytes quickly reorganize their cytoskeletons to elongate into a spindle shape, and the caliber of the lumen enlarges [14]. The muscle thickness of normal pulmonary arterioles falls considerably by two weeks of age [15], commensurate with the more gradual postnatal fall in pulmonary vascular resistance [16].

There are potential biological differences in congenital diaphragmatic hernia (CDH) an important contributor to aPH and one of the most frequent indications for Extracorporeal Membrane Oxygenation in the modern era. Both pulmonary hypoplasia and PH are associated with CDH and abnormal development begins early in pregnancy [17, 18]. Vascular growth is characterized by hyper-muscularization of larger vessels and neomuscularization of capillaries. The pathophysiological contributors are likely multiple, however, abnormal stabilization of new vascular endothelium by pericytes during the process of distal angiogenesis has been implicated as a potential contributor [19]. Abnormalities in circulating endothelin-1 and excessive endothelin A and B receptors have also been shown in human CDH [20, 21].

SECTION II: PHYSIOLOGY OF PULMONARY VASCULAR DYSREGULATION AND RIGHT VENTRICULAR DYSFUNCTION

- Activation of the pulmonary microvascular endothelium due to infective, immune complex, oxidative, micro-embolic, and other stressors is a key contributor to aPH. Pulmonary endothelial inflammation leads to pulmonary capillary occlusion and therefore hypoxia due to ventilation/perfusion mismatch.
- Pulmonary vascular disease leads to high right ventricular afterload, to which the RV is particularly vulnerable. Sustained exposure to high afterload leads to RV dilation and dysfunction which in turn perpetuates the cycle of low pulmonary blood flow, impaired left ventricular filling. Low cardiac output occurs and leads to further RV dysfunction due to impaired coronary perfusion and adverse ventricular interactions.

Acute PH in neonates and children is often triggered by a primary lung injury which then potentiates further dysregulation of the pulmonary vascular bed [22]. Hypoxic pulmonary vasoconstriction (HPV) is an important mechanism by which the lung and pulmonary vascular bed match ventilation (V) to perfusion (Q). It is biologically advantageous for well-aerated airspaces to receive the bulk of pulmonary blood flow via local vasodilation mediated by the presence of a high partial pressure of intra-alveolar oxygen. A low partial pressure of oxygen leads to cellular release of calcium in the pulmonary artery vascular smooth muscle cell primarily located at the entrance to the acinus and precipitates localized vasoconstriction (HPV) which diverts blood away from poorly ventilated areas. Among patients with lung disease, or other mediators known to exacerbate HPV [e.g., acidosis, hypothermia], alveolar hypoxia may contribute substantially to elevated PVR and aPH [23, 24]. A number of other factors contribute to vascular dysregulation that operate at different levels of the pulmonary vascular tree [Fig. 1]. Activation of the pulmonary microvascular endothelium is a key trigger in the cascade of pulmonary vascular dysregulation in the setting of aPH. Triggers such as microbes, immune complexes, drugs, toxins, reactive oxygen species, cytokines, and micro-emboli, induce pulmonary endothelial inflammation which in turn leads to pulmonary capillary occlusion through formation of intravascular microthrombi, fibrin, and intravascular sequestration of cells [22, 25]. The resulting hypoxia (due to V/Q mismatch from capillary occlusion) and increased local vasoactive mediators (e.g., endothelin-1 due to ongoing inflammation) leads to increased vascular tone in the pulmonary arterioles and veins [26, 27]. Furthermore, if the etiology is primary lung injury, then the consequent reduction in lung volume due to ongoing atelectasis, pulmonary edema, and high positive pressures on mechanical ventilation further constricts the pulmonary vessels [28]. Persistence of these pathobiological mechanisms lead to structural changes in the pulmonary vascular bed and/or elevated PVR. There is increased thickness of the existing pulmonary arterial smooth muscle layer, and muscularization of previously non-muscularized vessels. Moreover, the

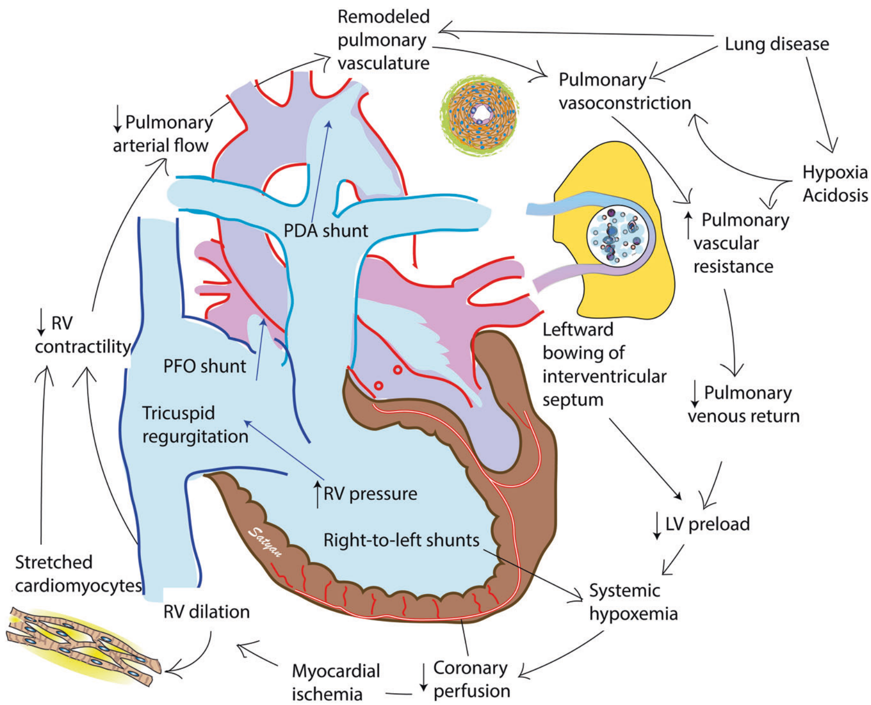

Fig. 1 The vicious cycle of neonatal acute pulmonary hypertension. Pulmonary disease and/or vascular remodeling leads to elevated pulmonary vascular resistance. Increased right ventricular afterload initially leads to right ventricular hypertrophy but subsequently causes dilation and poor contractility of the right ventricle (RV). Bowing of the interventricular septum and reduced pulmonary venous return lead to reduced left ventricular (LV) preload. Diminished systemic perfusion and right-to-left shunts at patent foramen ovale (PFO) and/or patent ductus arteriosus (PDA) contribute to systemic hypoxemia. Systemic hypoxemia and hypoperfusion reduce coronary perfusion pressure and exacerbate myocardial ischemia and ventricular dysfunction. Copyright Satyan Lakshminrusimha.

small pulmonary arteries, veins, and lymphatics develop fibrous intimal proliferative lesions [22, 25, 29].

Occasionally, pulmonary vein stenosis or elevated left atrial pressure may lead to aPH originating in the postcapillary region of the pulmonary vascular bed [30]. The latter may be observed in conditions with associated with systemic hypertension, left ventricular (LV) diastolic dysfunction, and mitral valve disease [30].

Cardiovascular implications of aPH

The pathophysiology of hemodynamic derangements in aPH represents the common final pathway of the effects of excessive right ventricular (RV) afterload (of which PVR is typically the primary contributor). Decreased pulmonary blood flow and RV function are the interrelated primary aberrations that culminate in circulatory instability. While the ellipse-shaped and thicker-walled left ventricle is well-equipped to function under conditions of increased afterload, the RV is thin walled and crescent-shaped, designed to function as a flow generator accommodating the entire systemic venous return to the heart [31]. In response to high PVR (and increased pulmonary artery pressure) the RV may adapt by increasing myocardial contractility (known as homeometric adaptation) [32], but if this adaptation fails, RV dilatation occurs, damaging the RV contractile apparatus by lengthening the individual sarcomeres beyond their optimal interactive capacity. The resultant decrease in RV stroke volume initiates a vicious cycle of reduced pulmonary blood flow, LV preload and output, systemic hypoxemia, and RV myocardial ischemia. RV dilatation causes bowing of the interventricular septum into the LV and in combination with reduced pulmonary venous return, results in impaired LV filling and function by disrupting the elliptical configuration that is optimal for LV myocardial performance. The resultant decrease in LV output potentiates further negative systolic interaction and RV-arterial uncoupling. In addition, change in septal configuration reduces effective 'systolic ventricular interdependence' which describes the typical transmission of force through the interventricular septum from the usually higher pressure left ventricle to the right ventricle accounting for up to 40% of the systolic force driving right ventricular ejection [33, 34].

Myocardial ischemia and hypoxia of the RV play an important role in the pathophysiology of circulatory failure in aPH; specifically, through inducing compromise to right coronary blood flow the ability of the RV to maintain contractile function under conditions of elevated afterload is impaired. In healthy neonates, the right ventricle receives blood flow in a phasic pattern throughout the entire cardiac cycle and has an appreciable flow reserve compared with the LV owing to higher coronary perfusion gradient (aortic pressure – RV pressure) as RV pressure is lower than LV pressure. In the setting of increased pulmonary artery pressure, however, the decrease in aortic pressure (related to reduced LV preload, function, and output) results in a reduction in right coronary perfusion pressure, which is further compromised by sustained increases in RV systolic and diastolic tissue pressures [35]. As systemic blood pressure decreases and RV pressure increases, RV ischemia ensues [36]. The critical importance of an imbalance between myocardial oxygen supply and demand in determining RV contractility is underscored by studies demonstrating reversal of the adverse effects of severe aPH on pulmonary blood flow and RV perfusion and oxidative metabolism after unselective-vasopressor induced increases in coronary artery perfusion pressure [37–39]. This decrease in effective coronary perfusion in the context of increasing myocardial oxygen consumption predisposes the right ventricle to ischemia and dysfunction. Acute RV failure as a result of this unfavorable milieu is the classic proximate cause of patient demise in a severe pulmonary hypertensive crisis [40, 41]. Restoration of systemic blood pressure to the physiological range appropriate for gestational age is likely to improve myocardial perfusion. Importantly, supra-physiological elevation of systemic blood pressure using vasopressor drugs may potentially place additional afterload on both ventricles which may be harmful in some situations [42].

Amongst neonates with circulatory instability, co-morbid neonatal lung disease (e.g., pneumonia or meconium aspiration syndrome) is negatively synergistic. The reduced PBF exacerbates pre-existing ventilation/perfusion mismatch, hypoxemia and acidosis, which induces further pulmonary vasoconstriction and may impair the response to pulmonary vasodilator therapies. Finally, the modulator role of the atrial/ductal shunts in aPH is an important consideration. In severe disease, pure pulmonary-to-systemic shunts may occur across the foramen ovale and ductus arteriosus, culminating in the systemic delivery of deoxygenated blood. While these 'right-to-left' shunts may potentiate ongoing hypoxemia by perpetuating the low pulmonary blood flow state, these shunts also reduce RV afterload by offloading the pulmonary circulation and support systemic blood flow (albeit with deoxygenated blood).

SECTION III: APPROACH TO CLINICAL DIAGNOSIS AND MONITORING

- Historical clues to conditions that predispose to aPH such as prolonged oligohydramnios, history of meconium-stained fluid, non-reassuring fetal status may help to raise the index of suspicion for aPH. Respiratory distress is characteristic of pulmonary parenchymal disease which may occur on its own or concurrently with aPH. Hypoxemia out of proportion to the magnitude of respiratory distress should raise concern for disordered pulmonary blood flow.
- A pre-ductal saturation probe should be placed on the right arm and a post-ductal probe on one foot. Higher pre-ductal SpO2 compared to the feet suggests aPH with an open ductus with right to left shunt, but also some forms of congenital heart disease. The lack of a pre/post ductal saturation difference does not exclude a diagnosis of aPH and is oftentimes seen in the most critically ill patients where the lack of a transitional shunt leads to impaired RV systolic performance.

Taking a good perinatal history can often provide clues to the underlying etiology of neonatal cyanosis (hypoxemia). Oligohydramnios (due to prolonged rupture of membranes since second trimester or renal anomalies) or polyhydramnios (due to fetal akinesia sequence) can be associated with lung hypoplasia, leading to hypoxemic respiratory failure [43, 44]. Prolonged rupture of membranes can also increase the likelihood of developing early-onset sepsis and congenital pneumonia [45, 46]. Maternal diabetes and multiple pregnancies can be associated with cardiomyopathy [47]. Antenatal exposure to selective serotonin reuptake inhibitors may be associated with pulmonary hypertension after birth [48]. Cardiac and respiratory compromise may follow perinatal asphyxia with hypoxic-ischemic insult to the myocardium [49].

A comprehensive, complete newborn examination is crucial to determine the cause of neonatal hypoxemia. Tachypnoea (respiratory rate >60/min), dyspnea, and grunting are highly suggestive of respiratory disease, while decreased respiratory drive (periodic breathing, frequent apnea) may be associated with prematurity, neurological conditions, or secondary to administered medications (e.g., toxic levels of phenobarbital in an infant treated for seizure). Though heart defects involving the right ventricular inflow (e.g., Ebstein's disease) or outflow (e.g., critical pulmonary stenosis) tract typically result in hypoxemia/cyanosis without dyspnea, some cardiac lesions may mimic respiratory diseases. For example, total anomalous pulmonary venous drainage (TAPVD) may present as HRF refractory to commonly used therapies, such as surfactants and inhaled nitric oxide (iNO); left ventricular outflow tract lesions (e.g., coarctation of aorta, hypoplastic left heart syndrome) can present with prominent respiratory distress and cyanosis at the time of ductal closure. The presence of systemic hypotension or discordance between pre- vs post-ductal arterial pressure (pre > post systolic arterial pressure in patients with open PDA) may also be seen in patients with aPH.

Chest radiography is an essential first line investigation tool in patients with concerns of HRF. Essential components in the review of chest radiography include lung volume (number of ribs), parenchyma, presence of air-leak, and endotracheal tube position, if present. Many of the radiological features, however, are non-specific. Decreased pulmonary vascular markings are suggestive of both severe aPH or significant right-sided outflow tract obstruction type of cardiac lesions. An enlarged heart may relate to dilation of one or both ventricles secondary to ventricular failure or may be seen in congenital heart disease [e.g., Ebstein's anomaly]. In contrast, a relatively small heart can be seen in ventilated infants with hyperventilation or decreased preload. Similarly, infants with obstructed infra-diaphragmatic anomalous pulmonary venous drainage may present in extremis with profound hypoxemia (pO2 < 25 mmHg), severe hemodynamic instability, and with radiographic evidence of dense perihilar pulmonary edema and a small heart. Fallot's Tetralogy and Transposition of Great Vessels can be associated with a "boot shaped" heart with an upturned apex and an "egg on a string" appearance respectively; the classic appearance of congenital heart disease on radiography, however, may be difficult to discern due to the thymic shadow.

Pulse oximetry for oxygen saturation is commonly utilized to continuously monitor the clinical status of critically ill neonates [Fig. 2]. Hypoxemia disproportionate to the severity of parenchymal disease on chest radiography may suggest aPH or a duct-dependent pulmonary blood flow lesion. Simultaneous measurement of pre-ductal (right hand) and post-ductal (foot) oxygen saturations may determine if there is any degree of right-to-left shunting at the ductal level in neonates complicated by aPH. The lack of a pre/post ductal saturation difference does not exclude a diagnosis of aPH and is oftentimes seen in the most critically ill patients where the lack of a transitional shunt leads to impaired RV systolic performance. As such, narrowing of pre- and post-ductal difference with a shock-like picture (poor lower limb perfusion, decreased urine output) can be a warning sign of closure of ductus in neonates with severe pulmonary hypertension & right heart failure or left-sided obstructive cardiac lesions.

Due to the non-specific nature of routine clinical assessment clinicians should adopt a low threshold for comprehensive echocardiography assessment, especially in the following clinical situations: [50, 51] (i) FiO2 > 0.35 despite appropriate respiratory support, (ii) Differential oxygenation in the pre and post ductal areas >5% by either pulse oximetry or arterial blood gases with assumed good signal, (iii) more than two unexplained desaturations (SpO2 < 85%) in 12 h period or (iv) unexplained hypotension or poor perfusion in a patient at high-risk for aPH [e.g., hypoxemic ischemic encephalopathy].

SECTION IV: STANDARDIZATION OF ECHOCARDIOGRAPHY ASSESSMENT IN APH

- Initial echocardiography should confirm the diagnosis of aPH, rule out important clinical mimics [e.g., pulmonary stenosis, anomalous pulmonary venous drainage, myocarditis, coarctation], and aim to quantify as best as feasible the pulmonary arterial pressure, RV performance, adequacy of pulmonary and systemic blood flow.
- Comprehensive and objective evaluation is desirable and may aid in longitudinal monitoring. The frequency of assessment should be determined by the acuity of the patient. Neonates

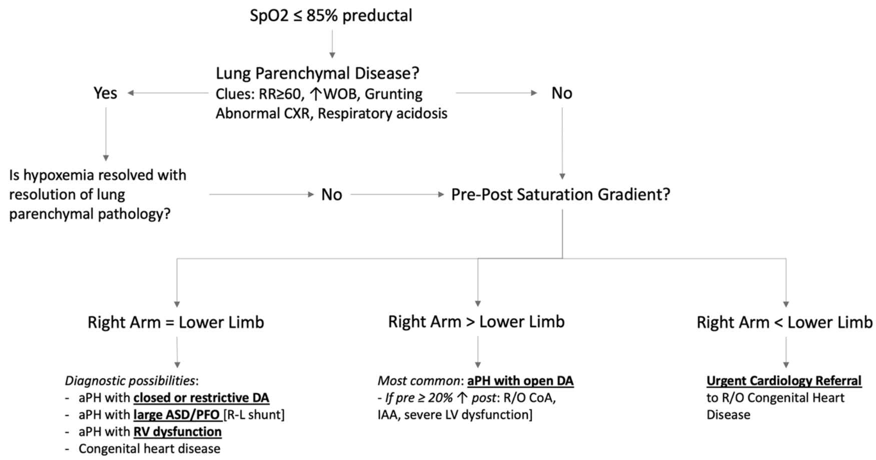

Fig. 2 Suggested approach to diagnosis of a neonate presenting with acute hypoxemia. Clinical features may aid in the pre-test probability of specific diagnosis prior to definitive echocardiography. SpO2 = oxygen saturation; RR = respiratory rate; WOB = work of breathing; CXR = chest x-ray; DA = ductus arteriosus; ASD = atrial septal defect; PFO = patent foramen ovale; aPH = acute pulmonary hypertension; RV = right ventricle; R/O = rule out; IAA = interrupted aortic arch; LV = left ventricle.

with refractory hypoxemia, hypotension, or poor perfusion may require multiple evaluations serially in the first 12–24h to monitor changing condition with therapeutic modifications or over time until stability is attained.

Initial echocardiography of patients with hypoxemic respiratory failure should be guided by clinical history [e.g., risk factors for aPH] and the severity of the clinical presentation. The most critical feature of this initial study is that imaging of sufficient quality to rule out anatomic heart disease is produced, regardless of the sonographers' background [e.g., pediatric, or adult cardiology, neonatology, critical care]. If the index of suspicion is high for anatomic disease, initial imaging should be performed by a pediatric cardiologist or their delegate [52, 53]. Important clinical mimics of aPH include anatomic limitations to pulmonary arterial [e.g., pulmonary stenosis] or venous blood flow [e.g. obstructed anomalous pulmonary venous drainage], severe left-sided disease which is either functional [e.g. myocarditis] or structural [e.g., aortic coarctation] and flow-driven pulmonary hypertension which may occur due to arteriovenous malformations. Once aPH is confirmed, quantification of pulmonary artery pressure (PAP), evaluation of the impact on right ventricular (RV) performance and on systemic blood flow (SBF). The interpretation of echocardiography information by hemodynamic consultants, with formal training in cardiopulmonary intensive care may be beneficial.

Contributors to acute hypoxemia in the transitional period are varied and all measures of PAP have different limitations [Table 1]. Importantly, PAP may be underestimated in the presence of RV dysfunction because all measures except ductal shunt rely on the RV to create a pressure gradient. Estimation of the difference between pulmonary and systemic pressure using the gradient across an unrestrictive PDA is validated [54, 55]. Exclusive right-to-left flow occurs when PAP is higher than systemic throughout the cardiac cycle. Among patients with bidirectional shunt, duration of right-to-left flow ≥60% systole is associated with supra-systemic PAP [54]. The presence of either pulmonary insufficiency or tricuspid regurgitation may be used to estimate the mean PAP and right ventricular systolic pressure (RVSp) respectively [56]. Subjective assessment of septal flattening has poor reliability [57]. Quantification of septal flattening using eccentricity index [58] has been shown to differentiate neonates with both acute [59] and chronic [60] PH from controls. The use of a trans-valvular regurgitant jet and transductal flow have the greatest repeatability [61]. Among infants and children, pulmonary artery acceleration time (PAAT) and an index of PAAT to right ventricular ejection time have been used to accurately estimate afterload [62]. Similarly, the pulmonary artery Doppler waveform shape correlates with RVSp in adults [56]. Neonatal data is limited, particularly at heart rates ≥135.

Elevated PAP may be well tolerated if sufficient pulmonary blood flow (PBF) is possible to maintain normal V/Q matching and cardiac output. The RV, however, is especially vulnerable to afterload [63, 64]. As previously described, sustained exposure to wall-stress may produce progressive dilation, dysfunction, and inadequate PBF. Assessment of RV size, systolic and diastolic performance are recommended using quantitative techniques with published normative data [65]. [Table 1] For output measurements, a zero angle of insonation, and precise placement of the sample volume at the level of the annulus is essential [66]. Pre-ductal LV output, which is the primary source of brain perfusion, reflects a combination of pulmonary venous return and net right-to-left trans-atrial flow. Effective post-ductal SBF is a combination of left ventricular output (LVO) and the net right-to-left ductal flow. In the presence of significant right-to-left ductal flow, therefore, true PBF may be lower than measured right ventricular output (RVO). This may be clinically desirable because ductal patency facilitates reduced RV afterload and prevents post-ductal ischemia.

Follow-up echocardiography should be determined by disease severity. Neonates with refractory hypoxemia, hypotension, or poor perfusion may require multiple evaluations over 12–24h until stability is attained. In contrast, for neonates with aPH and normal RV performance who have a good response to pulmonary vasodilator therapy, it is reasonable to initiate weaning after 12–24h of stability. One approach is to consider follow-up evaluations either if weaning is unsuccessful or after the discontinuation of therapy to confirm resolution of aPH [e.g., PAP < ½SAP]. Post-discharge follow-up is reasonable if aPH is ongoing or recurrent, requiring home oxygen, or if echocardiography has not normalized in hospital. It is, however, atypical for aPH to continue beyond the neonatal period, and therefore consideration of pulmonary vascular maldevelopment [e.g., alveolar capillary dysplasia] is reasonable for a multiphasic or protracted [≥2–4 weeks] course.

SECTION V: THERAPEUTIC APPROACH TO APH

- Appropriate lung recruitment and treatment of underlying pulmonary parenchymal disease are essential. Maintaining

Table 1. Echocardiography and hemodynamics indices in acute pulmonary hypertension.

Hemodynamic (Physiologic) parameter	**Clinical utility**	**Advantages**	**Limitations**
Measures of right ventricular function			
RVO [66] Right ventricular output; mL/min/kg	Surrogate marker of pulmonary blood flow and RV function	Published normal data which is useful to objectify pulmonary blood flow. Serial evaluations can greatly aid in following disease progression.	-Dependent on Doppler angle, position of sample volume and precise measurement -May over-estimate if dilated RVOT [e.g., prolonged PH]
RV FAC [65] Right ventricular fractional area change	Global measure of RV function.	Published normal data for the transitional period. Ejection fraction assessment of RV	-Needs high-quality imaging for endocardial definition -Influenced by septal motion in apical four-chamber
TAPSE; [114] tricuspid annular plane systolic excursion; mm	Measure of longitudinal RV contractility.	Published normal data. - Feasible and easy to measure - Independent of shunts and heart rate	-Angle dependent and assumes no regional wall motion defects -Exaggerated by severe TR
Measures of pulmonary pressure			
Physiologic parameter	Clinical utility	Advantages	Limitations
RVSP [114] Right ventricular systolic pressure; mmHg; utilizes tricuspid regurgitant jet	Direct measure of RV systolic pressure	Validated against invasive cardiac catheterization	-Tricuspid valve damage may cause TR with normal RVSP -Angle dependent and requires complete envelope -Absence of TR is common despite severe aPH
Subjective assessment of septal curvature	Used to judge relative RV vs LV pressure.	Easy to measure; universally available	Subjective and low inter-rater reliability, particularly for mild-moderate disease
Eccentricity index [58]		Reproducible, validated in neonates	Relative to LVSP, therefore may underestimate RVSP in systemic hypertension
PAP: AOP [114] Ratio of pulmonary artery to aortic pressure; utilizes Doppler of the ductus arteriosus.	Direct comparison of pulmonary to systemic pressure	Useful to qualify RVSP as sub systemic, supra systemic, or equal systemic	-Not possible without PDA and may be unreliable if PDA is restrictive, tortuous or S shaped. -Varies with ambient conditions [e.g., SpO_2, Ph]
Measures of pulmonary vascular resistance			
Physiologic parameter	Clinical utility	Advantages	Limitations
PAAT [115] Pulmonary artery acceleration time, msec	May be used to serially monitor changes in PVR with time or treatment	Easy to measure	Underestimated by RV dysfunction and unreliable in the presence of shunts. Single time-point measures Limited evidence in neonates.
RVET: PAAT [115] PAAT indexed to RV ejection time, msec			
Mid-systolic notching of PA Doppler waveform [116]			
Measures of systemic blood flow			
Physiologic parameter	Clinical utility	Advantages	Limitations
LVO [66] Left ventricular output; mL/min/kg	Surrogate marker of systemic blood flow and LV function	Published normal data. Serial evaluations can greatly aid in following disease progression.	Dependent on Doppler angle and position of sample volume PDA shunt dependent

SpO2 90–95% to achieve a PaO2 60–80 mmHg, arterial PCO2 35–45 mmHg and pH 7.3–7.45 are recommended to optimize the relationship between alveolar gas mixture and PVR. Alternative thresholds for patients with a diagnosis of congenital diaphragmatic hernia may be considered but are outside of the scope of this document. Once adequate lung recruitment has been achieved, inhaled nitric oxide is recommended as first line pulmonary vasodilator therapy.

- Non-selective vasopressors which augment both systemic and pulmonary vascular resistance [e.g., dopamine] should be avoided. Alternative vasopressors which have a lesser effect in the pulmonary vascular bed, such as vasopressin and possibly norepinephrine, may be more effective. Milrinone is an effective non-selective vasodilator, however, should be used with caution in patients with hypotension and those in whom clearance is questionable as it may reduce systemic blood pressure. Prostaglandin may be beneficial for patients with refractory disease both as a non-selective vasodilator and to maintain right to left ductal shunt as a pop-off for the RV.

The recommended therapeutic strategy among patients with aPH is one that addresses the primary problem of low pulmonary blood flow by reducing PVR, augmenting RV performance if required, normalizing systemic blood flow, and optimizing RV coronary perfusion pressure. Optimal management of lung parenchymal disease is a first principle in the management of

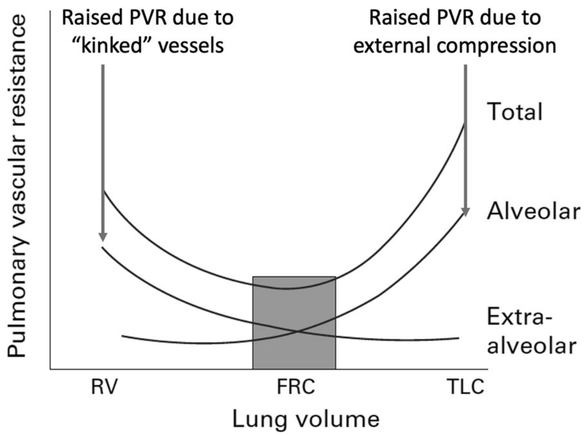

Fig. 3 Relationship between lung volume and pulmonary vascular resistance. At lung volumes below functional residual capacity, extra-alveolar pulmonary vessels are tortuous and therefore resistance to flow is high. Above functional residual capacity, as total lung volume is reached, alveolar vessels become compressed by airspace over distension. The balance of optimal pulmonary vascular resistance exists when lung volume is within the central range described in this figure.

aPH. Both over [67] and under-distension [68] of alveoli are associated with elevation in PVR and a lesser response to inhaled nitric oxide may be seen at the extremes of mean airway pressure [69] [Fig. 3]. Excessive intrathoracic pressure has been linked to a variety of negative hemodynamic consequences [70, 71]. Therefore, though no specific ventilation mode has demonstrated superiority, the importance of titrating ventilation to lung compliance cannot be underestimated. Patients with radiography features of respiratory distress syndrome or history of meconium aspiration syndrome [72, 73] may benefit from surfactant therapy. Caution is advised with administration, however, as surfactant treatment has been associated with transient airway obstruction which may acutely worsen V/Q matching [74]. Different strategies for deploying surfactant to prevent this consequence have not been systematically studied. Corticosteroids, such as hydrocortisone and/or dexamethasone are frequently used as adjunctive therapy for inflammatory etiologies such as pneumonia and should also be considered in meconium aspiration syndrome [75].

Optimal target oxygen saturation and/or arterial PO_2 has not been studied in depth in neonatal aPH. There is a wide range of practices with respect to approach to monitoring [e.g., saturation, PaO_2, oxygenation index] and specific targets [76, 77]. In term animal models, both alveolar hypoxia and hypoxemia stimulate pulmonary vasoconstriction [78] and a steep inverse relationship between pulmonary artery pressure and PaO_2 has been shown between 20 and 40 mmHg [79]. This relationship reaches a plateau at 50 mmHg suggesting that further augmentation of PaO_2 beyond 50 mmHg has no further benefit in reducing PVR and may contribute to creation of oxygen free-radicals which may, in turn, contribute to elevated PVR [80]. Based on the higher affinity of fetal hemoglobin for oxygen, maintaining a SpO_2 between 90–95% to achieve a PaO2 of 60–80mmHg is recommended [81, 82]. It is important to note that most PaO_2 measurements in this patient population are derived from umbilical arterial catheters which are located in the post-ductal circulation. Among patients with right-to-left ductal shunt, post-ductal PaO_2 may be substantially lower than pre-ductal values. A difference of 40–50 mmHg has been documented in neonatal patients with congenital diaphragmatic hernia and right-to-left ductal shunt [83]. Pre-ductal measurements of either PaO_2 or SpO_2 are, therefore, recommended to minimize oxygen toxicity, while supporting maximal pulmonary vasodilation. Alveolar carbon dioxide has a direct relationship with PVR [84, 85]. Historically, hyperventilation has been used as a treatment for aPH; [85] however, low CO_2 is associated with cerebral vasoconstriction. Maintaining CO_2 in the 35–45 mmHg range is recommended to balance the benefits in the pulmonary vascular bed with the risk of cerebral injury. In the setting of CDH, physiology a pre-ductal SpO2 between 85–95% and arterial pH > 7.25 are desirable to minimize excessive oxygen exposure or barotrauma in the pursuit of higher levels.

Once adequate lung recruitment has been achieved, inhaled nitric oxide (iNO) is recommended as first line as a pulmonary vasodilator due to improved oxygenation in approximately 50% of patients and a reduction in the need for extracorporeal membrane oxygenation [86]. Wherever possible, comprehensive echocardiography should be performed prior to use of iNO to minimize use in patients where pulmonary vasodilation may be harmful. For example, in patients with known congenital diaphragmatic hernia to exclude patients with pulmonary venous hypertensions secondary to isolated severe LV systolic or diastolic dysfunction and left atrial hypertension. The presence of a left-right atrial level shunt is pathognomonic of this phenotypic subset. In addition, iNO may have a negative effect in patients with pulmonary vein stenosis or obstructed pulmonary venous drainage. A detailed discussion regarding the applicability of trial findings to patients with congenital diaphragmatic hernia or extreme prematurity, however, is outside of the scope of this consensus statement. iNO acts by binding to soluble guanylate cyclase and thereby upregulating the production of cyclic guanylate-monophosphate (cGMP). cGMP leads to vasodilation by mediating multiple downstream effects including enhanced cytosolic calcium extrusion and modulation of several other vasoactive mediators [87]. Additionally, inhaled NO may counteract HPV, which may explain its role in pre-alveolar vasodilation and may be an important mechanism of action, particularly in situations of borderline alveolar hypoxemia or in the presence of pro-vasoconstricting stimuli such as acidosis [23]. It was previously hypothesized that the effects of iNO were exclusively in the pulmonary vascular bed, because NO binds rapidly to either heme iron nitrosyl-hemoglobin or oxyhemoglobin to form methemoglobin and nitrate, which are inactive. More recent evidence suggests, however, that there are two pathways by which stable NO compounds may be formed under normoxic conditions and thus facilitate a microvascular endocrine effect of iNO in the systemic circulation, although the lack of major impact on systemic vascular resistance would suggest otherwise. In the presence of oxyhemoglobin, a reaction that results in nitrosylation of a cysteine residue of hemoglobin to produce S-nitrosylated Hb is favored over the reactions resulting in NO inactivation previously described. Circulating red cells may, therefore, store NO which is released in areas of low partial pressures of oxygen and microvascular dilation may result. Additionally, nitrite anions generated from iNO may be recycled to form NO which may be released by deoxyhemoglobin in hypoxic tissues [88]. Other local non-cardiovascular effects of iNO should also be considered. iNO has been shown to reduce neutrophil-mediated oxidative stress and attenuate pro-coagulant factors in animal and adult models of acute lung injury. Exposure may also precipitate reactive nitrogen species production; specifically, excessive production of peroxynitrate promote vasoconstriction and contribute to surfactant dysfunction [89]. Additionally, prolonged exposure has been associated with impaired cellular respiration in animal models [90]. In summary, although iNO is a highly effective pulmonary vasodilator in many patients there remains gaps in understanding of the pulmonary specific effects of therapy warranting further investigation.

Inhaled NO alone may be sufficient to improve pre-ductal systolic hypotension by improving pulmonary blood flow and reducing RV afterload [88]. Therefore, iNO responsiveness should be determined not only on the basis of improved metrics of systemic oxygenation but also on improvement in hemodynamics, specifically systolic arterial pressure. Many centers have iNO

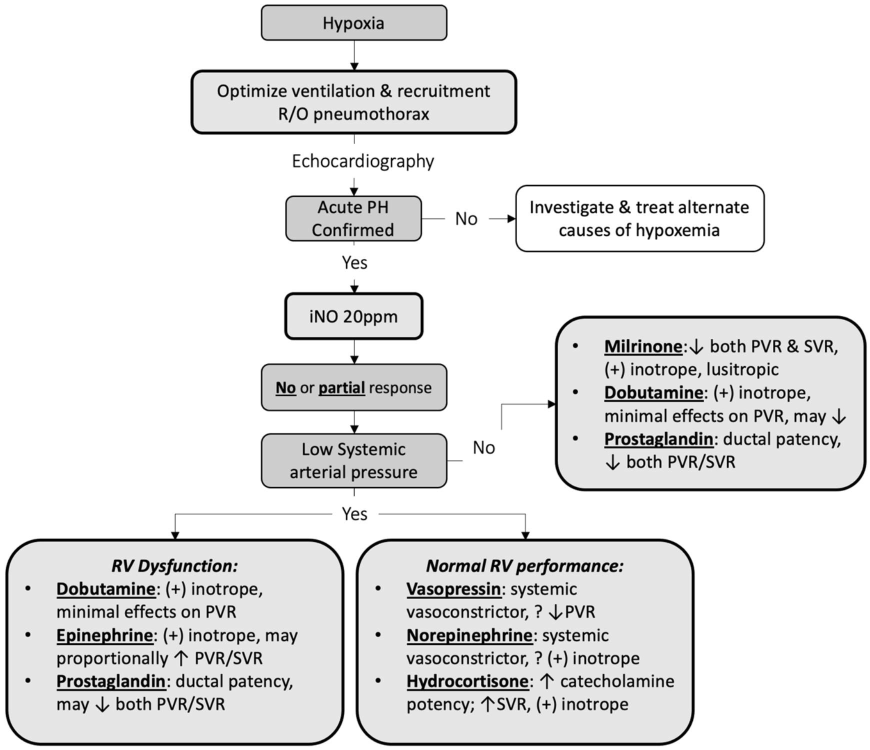

Fig. 4 Therapeutic strategies for the treatment of pulmonary hypertension refractory to inhaled nitric oxide therapy. Reduction of pulmonary vascular resistance is a fundamental principle which can be accomplished using vasodilator therapies such as milrinone or sildenafil. Both of these medications are non-selective vasodilators, however, and therefore maintenance of normal blood pressure using vasopressin or norepinephrine may be required. Support for right ventricular function is also important. The severely dysfunctional RV may be dependent on high central venous pressure for forward flow as in Fontan circulation. Prostaglandin may improve RV performance by allowing an alternative pathway to flow right to left via the ductus arteriosus into the lower resistance, and therefore, lower afterload systemic circulation.

Stewardship programs designed to optimize the balance of patient safety and medication cost [91, 92]. The weaning protocols typically used suggest that once stability is attained on a target dose of 20ppm, a regular weaning plan which reduces iNO dose every 1–2h to either the lowest tolerated dose or off, are an important key to appropriate resource utilization [92]. Given iNO's activity as a modulator of pulmonary blood flow, these weaning guidelines should consider, however, not only changes in oxygen requirement but also hemodynamic changes when evaluating the success of weaning. For patients with severe cardiovascular disease requiring other vasopressor or inotropes that impact the pulmonary vascular bed, or for those with known RV dysfunction the merits of weaning should be considered in the full clinical context. For example, a patient with severe RV dysfunction requiring agents to modify heart function and blood flow is probably not an ideal candidate for weaning even if requiring less oxygen than the protocolized set point.

For patients who have hypotension which is severe, refractory to iNO, or also has a diastolic component, vasopressors may be indicated to improve coronary perfusion pressure and/or RV preload. Non-selective vasopressor drugs which augment both systemic and pulmonary vascular resistance, of which dopamine is the most commonly used [76], should be avoided. The use of dopamine to augment systemic blood pressure in this patient population is commonly associated with progressive hypoxia. In addition to the negative impact on RV afterload described previously, dopamine may exacerbate tachycardia, which may adversely impact contractility via the force-frequency relationship, particularly among patients with pre-existing cardiac dysfunction [93, 94]. Vasopressin, an endogenously produced hormone that acts as a systemic vasoconstrictor primarily in the skin and splanchnic circulation, improves heart function in a piglet model of aPH [95] and has been used successfully to treat neonatal patients with aPH [96, 97]. Additionally, in the pulmonary circulation vasopressin binds to endothelial receptors to potentiate nitric oxide release, which may theoretically be beneficial. There are limited data regarding the efficacy of norepinephrine; however, animal models suggest benefit on PVR and RV performance [37, 98] and small studies suggest improved hemodynamics among this patient population [39]. There is, however, experimental evidence from an ovine model of PH that norepinephrine may act as a potent pulmonary vasoconstrictor when co-administered with 100% oxygen [99]. This effect is thought to be modulated by oxygen free-radicals of which superoxide is the most important.

Milrinone, a phosphodiesterase 3 inhibitor, is a drug that has been shown to be synergistic with iNO in adult animal models [100]. Its primary mechanism is to increase availability of cAMP and is useful as an adjunctive pulmonary vasodilator [101]. Positive inotropy via sarcoplasmic reticulum mediated modulation of calcium is another potentially beneficial property of milrinone, however, it may be less important in neonates for whom the development of the sarcoplasmic reticulum may be incomplete [102, 103]. An important caveat to milrinone use is its vasodilating role in the systemic circulation. Milrinone has been associated with improved blood pressure in some studies of aPH [104], likely due to improved pulmonary blood flow, however, as a non-specific vasodilator it is also associated with a reduction in diastolic pressure. Caution should be exercised among patients who are already hypotensive, particularly those with hypoxic-ischemic encephalopathy undergoing therapeutic hypothermia when milrinone metabolism and clearance can be expected to be impaired [105]. Outside of a single pharmacologic study, there is little data to guide the use of a loading dose of milrinone in term patients with aPH [106]. In theory, a loading dose should be tolerated provided adequate systemic arterial pressure is present prior to its initiation, however, given milrinone's long half-life and potential issues with clearance, the risks of prolonged systemic hypotension may outweigh the benefits in many populations, and caution is advised. In the absence of a loading dose the time to achieve a therapeutic level of milrinone is likely to be longer. There are also limited data regarding alternative systemically delivered pulmonary vasodilators. Sildenafil, a phosphodiesterase 5 inhibitor, has a theoretical advantage over iNO in that it may be available in low resource settings where iNO is not. As such, it has been studied in a limited number of patients compared to placebo, alternative vasodilators, or iNO. In settings where iNO is not available, sildenafil has some data to suggest that both alone [107] or in combination with milrinone [108] it may provide benefit. Unlike iNO, however, sildenafil is delivered systemically and may therefore exacerbate intrapulmonary shunt by reducing the efficacy of HPV and so its use as an alternative when both agents are available should be carefully considered. Prostacylin analogues such as epoprostenol, treprostinil and iloprost have biologically logical benefit in that, similar to milrinone, they target the cyclic AMP pathway [109]. There are a variety of different preparations which may be given via inhalation, intravenously or subcutaneously, however, there is limited data on their use in term patients with aPH and the systemic preparations have, similar to milrinone, been associated with systemic hypotension and carry a similar risk of impaired VQ mismatch as sildenafil.

Dobutamine, which is a positive inotrope with minimal effect on PVR [110], and epinephrine which increases PVR and SVR proportionally [111] are alternate positive inotropic agents which could be considered in different situations [Fig. 4]. Finally, prostaglandin infusion may have a dual role among patients with refractory aPH. First, prostaglandin is a non-specific vasodilator and may independently reduce PVR [112]. Second, maintaining ductal patency reduces RV afterload by permitting run-off into the relatively lower resistance systemic circulation and, as an added benefit, right-to-left ductal shunt may prevent ischemia of post-ductal, and at times pre-ductal organs when left ventricular output is low [113] [Fig. 4].

In summary, acute pulmonary hypertension in patients with HRF represents a complex physiologic condition of diverse etiology and variable impact. It is incumbent of neonatal intensivists to develop an enhanced mechanistic understanding of the biologic modulators of this condition and its impact on both pulmonary blood flow and heart function. Optimization of lung recruitment and the provision of disease-specific treatments e.g., surfactant, antibiotics are essential intensive care pre-requisites. Pulmonary blood flow is modulated by RV function, PVR, and transitional shunts, although the relative contribution of each is likely to vary between patients or according to concurrent competing interventions e.g., mechanical ventilation. Comprehensive targeted neonatal echocardiography facilitates enhanced diagnostic precision and the selection of treatments which are more physiologically relevant. The latter is critically important in patients who are refractory to standard of care treatments such as mechanical ventilation or inhaled nitric oxide.

REFERENCES

1. Lakshminrusimha S, Steinhorn RH. Pulmonary vascular biology during neonatal transition. Clin Perinatol. 1999;26:601–19.
2. Haworth SG. Development of the normal and hypertensive pulmonary vasculature. Exp Physiol. 1995;80:843–53.
3. Schittny JC. Development of the lung. Cell Tissue Res. 2017;367:427–44.
4. Hall SM, Hislop AA, Pierce CM, Haworth SG. Prenatal origins of human intrapulmonary arteries: formation and smooth muscle maturation. Am J Respir Cell Mol Biol. 2000;23:194–203.
5. Hislop AA. Airway and blood vessel interaction during lung development. J Anat. 2002;201:325–34.
6. Belik J, Halayko AJ, Rao K, Stephens NL. Pulmonary and systemic vascular smooth muscle mechanical characteristics in newborn sheep. Am J Physiol. 1992;263:H881–6.
7. Lakshminrusimha S. The pulmonary circulation in neonatal respiratory failure. Clin Perinatol. 2012;39:655–83.
8. Lewis AB, Heymann MA, Rudolph AM. Gestational changes in pulmonary vascular responses in fetal lambs in utero. Circ Res. 1976;39:536–41.
9. Morin FC 3rd, Egan EA. Pulmonary hemodynamics in fetal lambs during development at normal and increased oxygen tension. J Appl Physiol (1985). 1992;73:213–8.
10. Storme L, Rairigh RL, Parker TA, Kinsella JP, Abman SH. In vivo evidence for a myogenic response in the fetal pulmonary circulation. Pediatr Res. 1999;45:425–31.
11. Mathew R, Altura BM. Physiology and pathophysiology of pulmonary circulation. Microcirc Endothelium Lymphatics. 1990;6:211–52.
12. Fediuk J, Dakshinamurti S. A role for actin polymerization in persistent pulmonary hypertension of the newborn. Can J Physiol Pharm. 2015;93:185–94.
13. Basu S, Datta BN, Khandelwal N. Morphologic changes in pulmonary vasculature with arteriographic correlation. Angiology. 1996;47:375–80.
14. Allen K, Haworth SG. Human postnatal pulmonary arterial remodeling. Ultrastructural studies of smooth muscle cell and connective tissue maturation. Lab Invest. 1988;59:702–9.
15. Michel RP, Gordon JB, Chu K. Development of the pulmonary vasculature in newborn lambs: structure-function relationships. J Appl Physiol (1985). 1991;70:1255–64.
16. Haworth SG, Hislop AA. Adaptation of the pulmonary circulation to extra-uterine life in the pig and its relevance to the human infant. Cardiovasc Res. 1981;15:108–19.
17. Harting MT. Congenital diaphragmatic hernia-associated pulmonary hypertension. Semin Pediatr Surg. 2017;26:147–53.
18. Sluiter I, van der Horst I, van der Voorn P, Boerema-de Munck A, Buscop-van Kempen M, de Krijger R, et al. Premature differentiation of vascular smooth muscle cells in human congenital diaphragmatic hernia. Exp Mol Pathol. 2013;94:195–202.
19. Mous DS, Kool HM, Wijnen R, Tibboel D, Rottier RJ. Pulmonary vascular development in congenital diaphragmatic hernia. Eur Respir Rev. 2018;27:170104 https://doi.org/10.1183/16000617.0104-2017.
20. de Lagausie P, de Buys-Roessingh A, Ferkdadji L, Saada J, Aisenfisz S, Martinez-Vinson C, et al. Endothelin receptor expression in human lungs of newborns with congenital diaphragmatic hernia. J Pathol. 2005;205:112–8.
21. Keller RL, Tacy TA, Hendricks-Munoz K, Xu J, Moon-Grady AJ, Neuhaus J, et al. Congenital diaphragmatic hernia: endothelin-1, pulmonary hypertension, and disease severity. Am J Respir Crit Care Med. 2010;182:555–61.
22. Price LC, McAuley DF, Marino PS, Finney SJ, Griffiths MJ, Wort SJ. Pathophysiology of pulmonary hypertension in acute lung injury. Am J Physiol Lung Cell Mol Physiol. 2012;302:L803–15.
23. Dunham-Snary KJ, Wu D, Sykes EA, Thakrar A, Parlow LRG, Mewburn JD, et al. Hypoxic pulmonary vasoconstriction: from molecular mechanisms to medicine. Chest 2017;151:181–92.
24. Sommer N, Strielkov I, Pak O, Weissmann N. Oxygen sensing and signal transduction in hypoxic pulmonary vasoconstriction. Eur Respir J. 2016;47:288–303.
25. Zapol WM, Kobayashi K, Snider MT, Greene R, Laver MB. Vascular obstruction causes pulmonary hypertension in severe acute respiratory failure. Chest. 1977;71:306–7.
26. Brimioulle S, LeJeune P, Naeije R. Effects of hypoxic pulmonary vasoconstriction on pulmonary gas exchange. J Appl Physiol (1985). 1996;81:1535–43.
27. Rossi P, Persson B, Boels PJ, Arner A, Weitzberg E, Oldner A. Endotoxemic pulmonary hypertension is largely mediated by endothelin-induced venous constriction. Intensive Care Med. 2008;34:873–80.
28. Kuebler WM, Ying X, Singh B, Issekutz AC, Bhattacharya J. Pressure is proinflammatory in lung venular capillaries. J Clin Invest. 1999;104:495–502.
29. Tomashefski JF Jr., Davies P, Boggis C, Greene R, Zapol WM, Reid LM. The pulmonary vascular lesions of the adult respiratory distress syndrome. Am J Pathol. 1983;112:112–26.
30. Kulik TJ. Pulmonary hypertension caused by pulmonary venous hypertension. Pulm circulation. 2014;4:581–95.
31. Haddad F, Hunt SA, Rosenthal DN, Murphy DJ. Right ventricular function in cardiovascular disease, part I: anatomy, physiology, aging, and functional assessment of the right ventricle. Circulation. 2008;117:1436–48.
32. Sarnoff SJ, Mitchell JH, Gilmore JP, Remensnyder JP. Homeometric autoregulation in the heart. Circ Res. 1960;8:1077–91.
33. Brinker JA, Weiss JL, Lappe DL, Rabson JL, Summer WR, Permutt S, et al. Leftward septal displacement during right ventricular loading in man. Circulation. 1980;61:626–33.

34. Bronicki RA, Anas NG. Cardiopulmonary interaction. Pediatr Crit Care Med. 2009;10:313–22.
35. Brooks H, Kirk ES, Vokonas PS, Urschel CW, Sonnenblick EH. Performance of the right ventricle under stress: relation to right coronary flow. J Clin Invest. 1971;50:2176–83.
36. Naeije R, Manes A. The right ventricle in pulmonary arterial hypertension. Eur Respir Rev. 2014;23:476–87.
37. Hirsch LJ, Rooney MW, Wat SS, Kleinmann B, Mathru M. Norepinephrine and phenylephrine effects on right ventricular function in experimental canine pulmonary embolism. Chest. 1991;100:796–801.
38. Schreuder WO, Schneider AJ, Groeneveld AB, Thijs LG. Effect of dopamine vs norepinephrine on hemodynamics in septic shock. Emphasis on right ventricular performance. Chest. 1989;95:1282–8.
39. Tourneux P, Rakza T, Bouissou A, Krim G, Storme L. Pulmonary circulatory effects of norepinephrine in newborn infants with persistent pulmonary hypertension. J Pediatr. 2008;153:345–9.
40. Vlahakes GJ, Turley K, Hoffman JI. The pathophysiology of failure in acute right ventricular hypertension: hemodynamic and biochemical correlations. Circulation 1981;63:87–95.
41. Mebazaa A, Karpati P, Renaud E, Algotsson L. Acute right ventricular failure–from pathophysiology to new treatments. Intensive Care Med. 2004;30:185–96.
42. Cheung PY, Barrington K. The effects of dopamine and epinephrine on hemodynamics and oxygen metabolism in hypoxic anesthetized piglets. Crit Care. 2001;5:158–66.
43. Spong CY. Preterm premature rupture of the fetal membranes complicated by oligohydramnios. Clin Perinatol. 2001;28:753–9 vi.
44. Niles KM, Blaser S, Shannon P, Chitayat D. Fetal arthrogryposis multiplex congenita/fetal akinesia deformation sequence (FADS) - Aetiology, diagnosis and management. Prenat Diagn. 2019;39:720–31.
45. Waters TP, Mercer B. Preterm PROM: prediction, prevention, principles. Clin Obstet Gynecol. 2011;54:307–12.
46. Lindenskov PH, Castellheim A, Saugstad OD, Mollnes TE. Meconium aspiration syndrome: possible pathophysiological mechanisms and future potential therapies. Neonatology. 2015;107:225–30.
47. Pedra SR, Smallhorn JF, Ryan G, Chitayat D, Taylor GP, Khan R, et al. Fetal cardiomyopathies: pathogenic mechanisms, hemodynamic findings, and clinical outcome. Circulation. 2002;106:585–91.
48. Huybrechts KF, Bateman BT, Palmsten K, Desai RJ, Patorno E, Gopalakrishnan C, et al. Antidepressant use late in pregnancy and risk of persistent pulmonary hypertension of the newborn. JAMA. 2015;313:2142–51.
49. Barberi I, Calabro MP, Cordaro S, Gitto E, Sottile A, Prudente D, et al. Myocardial ischaemia in neonates with perinatal asphyxia. Electrocardiographic, echocardiographic and enzymatic correlations. Eur J Pediatr. 1999;158:742–7.
50. Lakshminrusimha S, Konduri GG, Steinhorn RH. Considerations in the management of hypoxemic respiratory failure and persistent pulmonary hypertension in term and late preterm neonates. J Perinatol. 2016;36:S12–9.
51. Lakshminrusimha S, Saugstad OD. The fetal circulation, pathophysiology of hypoxemic respiratory failure and pulmonary hypertension in neonates, and the role of oxygen therapy. J Perinatol. 2016;36:S3–s11.
52. Corredera A, Rodriguez MJ, Arevalo P, Llorente B, Moro M, Arruza L. [Functional echocardiography in neonatal intensive care: 1 year experience in a unit in Spain]. Pediatr (Barc). 2014;81:167–73.
53. Mertens L, Seri I, Marek J, Arlettaz R, Barker P, McNamara P, et al. Targeted neonatal echocardiography in the neonatal intensive care unit: practice guidelines and recommendations for training. Eur J Echocardiogr. 2011;12:715–36.
54. Musewe NN, Poppe D, Smallhorn JF, Hellman J, Whyte H, Smith B, et al. Doppler echocardiographic measurement of pulmonary artery pressure from ductal Doppler velocities in the newborn. J Am Coll Cardiol. 1990;15:446–56.
55. Musewe NN, Smallhorn JF, Benson LN, Burrows PE, Freedom RM. Validation of Doppler-derived pulmonary arterial pressure in patients with ductus arteriosus under different hemodynamic states. Circulation. 1987;76:1081–91.
56. Parasuraman S, Walker S, Loudon BL, Gollop ND, Wilson AM, Lowery C, et al. Assessment of pulmonary artery pressure by echocardiography-A comprehensive review. Int J Cardiol Heart Vasc. 2016;12:45–51.
57. Smith A, Purna JR, Castaldo MP, Ibarra-Rios D, Giesinger RE, Rios DR, et al. Accuracy and reliability of qualitative echocardiography assessment of right ventricular size and function in neonates. Echocardiography. 2019;36:1346–52.
58. Ryan T, Petrovic O, Dillon JC, Feigenbaum H, Conley MJ, Armstrong WF. An echocardiographic index for separation of right ventricular volume and pressure overload. J Am Coll Cardiol. 1985;5:918–27.
59. Aggarwal S, Natarajan G. Echocardiographic correlates of persistent pulmonary hypertension of the newborn. Early Hum Dev. 2015;91:285–9.
60. McCrary AW, Malowitz JR, Hornick CP, Hill KD, Cotten CM, Tatum GH, et al. Differences in eccentricity index and systolic-diastolic ratio in extremely low-birth-weight infants with bronchopulmonary dysplasia at risk of pulmonary hypertension. Am J Perinatol. 2016;33:57–62.
61. Skinner JR, Boys RJ, Heads A, Hey EN, Hunter S. Estimation of pulmonary arterial pressure in the newborn: study of the repeatability of four Doppler echocardiographic techniques. Pediatr Cardiol. 1996;17:360–9.
62. Levy PT, Patel MD, Singh GK. Reply. J Pediatr. 2018;202:336–7.
63. Reller MD, Morton MJ, Reid DL, Thornburg KL. Fetal lamb ventricles respond differently to filling and arterial pressures and to in utero ventilation. Pediatr Res. 1987;22:621–6.
64. Pinson CW, Morton MJ, Thornburg KL. An anatomic basis for fetal right ventricular dominance and arterial pressure sensitivity. J Dev Physiol. 1987;9:253–69.
65. Jain A, Mohamed A, El-Khuffash A, Connelly KA, Dallaire F, Jankov RP, et al. A comprehensive echocardiographic protocol for assessing neonatal right ventricular dimensions and function in the transitional period: normative data and z scores. J Am Soc Echocardiogr. 2014;27:1293–304.
66. Slama M, Susic D, Varagic J, Ahn J, Frohlich ED. Echocardiographic measurement of cardiac output in rats. Am J Physiol Heart Circ Physiol. 2003;284:H691–7.
67. Jardin F, Vieillard-Baron A. Right ventricular function and positive pressure ventilation in clinical practice: from hemodynamic subsets to respirator settings. Intensive Care Med. 2003;29:1426–34.
68. Creamer KM, McCloud LL, Fisher LE, Ehrhart IC. Ventilation above closing volume reduces pulmonary vascular resistance hysteresis. Am J Respir Crit Care Med. 1998;158:1114–9.
69. Hoffman GM, Nelin LD. Mean airway pressure and response to inhaled nitric oxide in neonatal and pediatric patients. Lung. 2005;183:441–53.
70. Whittenberger JL, Mc GM, Berglund E, Borst HG. Influence of state of inflation of the lung on pulmonary vascular resistance. J Appl Physiol. 1960;15:878–82.
71. Reller MD, Donovan EF, Kotagal UR. Influence of airway pressure waveform on cardiac output during positive pressure ventilation of healthy newborn dogs. Pediatr Res. 1985;19:337–41.
72. El Shahed AI, Dargaville PA, Ohlsson A, Soll R. Surfactant for meconium aspiration syndrome in term and late preterm infants. The Cochrane database of systematic reviews. 2014;2014:Cd002054.
73. Natarajan CK, Sankar MJ, Jain K, Agarwal R, Paul VK. Surfactant therapy and antibiotics in neonates with meconium aspiration syndrome: a systematic review and meta-analysis. J Perinatol. 2016;36:S49–54.
74. Tarawneh A, Kaczmarek J, Bottino MN, Sant'anna GM. Severe airway obstruction during surfactant administration using a standardized protocol: a prospective, observational study. J Perinatol. 2012;32:270–5.
75. Mokra D, Mokry J. Glucocorticoids in the treatment of neonatal meconium aspiration syndrome. Eur J Pediatr. 2011;170:1495–505.
76. Nakwan N, Chaiwiriyawong P. An international survey on persistent pulmonary hypertension of the newborn: a need for an evidence-based management. J neonatal-Perinat Med. 2016;9:243–50.
77. Alapati D, Jassar R, Shaffer TH. Management of supplemental oxygen for infants with persistent pulmonary hypertension of newborn: a survey. Am J Perinatol. 2017;34:276–82.
78. Hauge A. Hypoxia and pulmonary vascular resistance. The relative effects of pulmonary arterial and alveolar PO2. Acta Physiol Scand. 1969;76:121–30.
79. Rudolph AM, Yuan S. Response of the pulmonary vasculature to hypoxia and H + ion concentration changes. J Clin Invest. 1966;45:399–411.
80. Belik J, Jankov RP, Pan J, Tanswell AK. Peroxynitrite inhibits relaxation and induces pulmonary artery muscle contraction in the newborn rat. Free Radic Biol Med. 2004;37:1384–92.
81. Stadler SS, Macri CJ, Kopelman JN, Mitchell A, Chakraborty PK, Satin AJ. Effect of meconium on the hemoglobin-oxygen association curve. J Matern Fetal Med. 1999;8:253–5.
82. Walsh BK. Oxygen Administration. Neonatal and Pediatric Respiratory Care. St. Louis, Missouri: Elsevier Health Sciences; 2014. p. 148–62.
83. Gien J, Kinsella JP. Differences in preductal and postductal arterial blood gas measurements in infants with severe congenital diaphragmatic hernia. Arch Dis Child Fetal Neonatal Ed. 2016;101:F314–8.
84. Mora GA, Pizarro C, Jacobs ML, Norwood WI. Experimental model of single ventricle. Influence of carbon dioxide on pulmonary vascular dynamics. Circulation. 1994;90:Ii43–6.
85. Peckham GJ, Fox WW. Physiologic factors affecting pulmonary artery pressure in infants with persistent pulmonary hypertension. J Pediatr. 1978;93:1005–10.
86. Barrington KJ, Finer N, Pennaforte T, Altit G. Nitric oxide for respiratory failure in infants born at or near term. Cochrane database Syst Rev. 2017;1:Cd000399.
87. Mónica FZ, Bian K, Murad F. The endothelium-dependent nitric oxide-cGMP pathway. Adv Pharm. 2016;77:1–27.
88. Creagh-Brown BC, Griffiths MJ, Evans TW. Bench-to-bedside review: Inhaled nitric oxide therapy in adults. Crit Care. 2009;13:221.

89. Robbins CG, Davis JM, Merritt TA, Amirkhanian JD, Sahgal N, Morin FC 3rd, et al. Combined effects of nitric oxide and hyperoxia on surfactant function and pulmonary inflammation. Am J Physiol. 1995;269:L545–50.
90. Clementi E, Brown GC, Feelisch M, Moncada S. Persistent inhibition of cell respiration by nitric oxide: crucial role of S-nitrosylation of mitochondrial complex I and protective action of glutathione. Proc Natl Acad Sci USA. 1998;95:7631–6.
91. Ahearn J, Panda M, Carlisle H, Chaudhari T. Impact of inhaled nitric oxide stewardship programme in a neonatal intensive care unit. J Paediatr Child Health. 2020;56:265–71.
92. Elmekkawi A, More K, Shea J, Sperling C, Da Silva Z, Finelli M, et al. Impact of stewardship on inhaled nitric oxide utilization in a neonatal ICU. Hosp Pediatr. 2016;6:607–15.
93. Hasenfuss G, Schillinger W, Lehnart SE, Preuss M, Pieske B, Maier LS, et al. Relationship between Na+-Ca2+-exchanger protein levels and diastolic function of failing human myocardium. Circulation. 1999;99:641–8.
94. Pieske B, Trost S, Schutt K, Minami K, Just H, Hasenfuss G. Influence of forskolin on the force-frequency behavior in nonfailing and end-stage failing human myocardium. Basic Res Cardiol. 1998;93:66–75.
95. Pelletier JS, LaBossiere J, Dicken B, Gill RS, Sergi C, Tahbaz N, et al. Low-dose vasopressin improves cardiac function in newborn piglets with acute hypoxia-reoxygenation. Shock (Augusta, Ga). 2013;40:320–6.
96. Mohamed A, Nasef N, Shah V, McNamara PJ. Vasopressin as a rescue therapy for refractory pulmonary hypertension in neonates: case series. Pediatr Crit Care Med. 2014;15:148–54.
97. Shivananda S, Ahliwahlia L, Kluckow M, Luc J, Jankov R, McNamara P. Variation in the management of persistent pulmonary hypertension of the newborn: a survey of physicians in Canada, Australia, and New Zealand. Am J Perinatol. 2012;29:519–26.
98. Tulloh RM, Dyamenahalli U, Stuart-Smith K, Haworth SG. Adrenoceptor-stimulated endothelium-dependent relaxation in porcine intrapulmonary arteries. Pulm Pharm. 1994;7:299–303.
99. Lakshminrusimha S, Russell JA, Wedgwood S, Gugino SF, Kazzaz JA, Davis JM, et al. Superoxide dismutase improves oxygenation and reduces oxidation in neonatal pulmonary hypertension. Am J respiratory Crit care Med. 2006;174:1370–7.
100. Deb B, Bradford K, Pearl RG. Additive effects of inhaled nitric oxide and intravenous milrinone in experimental pulmonary hypertension. Crit Care Med. 2000;28:795–9.
101. Rao S, Bartle D, Patole S. Current and future therapeutic options for persistent pulmonary hypertension in the newborn. Expert Rev Cardiovasc Ther. 2010;8:845–62.
102. Ogawa S, Nakanishi T, Kamata K, Takao A. Effect of milrinone on myocardial mechanical function and cyclic AMP content in the fetal rabbit. Pediatr Res. 1987;22:282–5.
103. Sys SU, Goenen MJ, Chalant CH, Brutsaert DL. Inotropic effects of amrinone and milrinone on contraction and relaxation of isolated cardiac muscle. Circulation. 1986;73:Iii25–35.
104. McNamara PJ, Laique F, Muang-In S, Whyte HE. Milrinone improves oxygenation in neonates with severe persistent pulmonary hypertension of the newborn. J Crit Care. 2006;21:217–22.
105. Bischoff AR, Habib S, McNamara PJ, Giesinger RE. Hemodynamic response to milrinone for refractory hypoxemia during therapeutic hypothermia for neonatal hypoxic ischemic encephalopathy. J Perinatol. 2021;41:2345–54.
106. McNamara PJ, Shivananda SP, Sahni M, Freeman D, Taddio A. Pharmacology of milrinone in neonates with persistent pulmonary hypertension of the newborn and suboptimal response to inhaled nitric oxide. Pediatr Crit Care Med. 2013;14:74–84.
107. Kelly LE, Ohlsson A, Shah PS. Sildenafil for pulmonary hypertension in neonates. Cochrane database Syst Rev. 2017;8:Cd005494.
108. El-Ghandour M, Hammad B, Ghanem M, Antonios MAM. Efficacy of milrinone plus sildenafil in the treatment of neonates with persistent pulmonary hypertension in resource-limited settings: results of a randomized, double-blind trial. Paediatr Drugs. 2020;22:685–93.
109. Lakshminrusimha S, Mathew B, Leach CL. Pharmacologic strategies in neonatal pulmonary hypertension other than nitric oxide. Semin Perinatol. 2016;40:160–73.
110. Ruffolo RR Jr. The pharmacology of dobutamine. Am J Med Sci. 1987;294:244–8.
111. Barrington KJ, Finer NN, Chan WK. A blind, randomized comparison of the circulatory effects of dopamine and epinephrine infusions in the newborn piglet during normoxia and hypoxia. Crit Care Med. 1995;23:740–8.
112. Cassin S, Tyler T, Leffler C, Wallis R. Pulmonary and systemic vascular responses of perinatal goats to prostaglandins E1 and E2. Am J Physiol. 1979;236:H828–32.
113. Graham EM, Bradley SM, Atz AM. Preoperative management of hypoplastic left heart syndrome. Expert Opin Pharmacother. 2005;6:687–93.
114. Badano LP, Ginghina C, Easaw J, Muraru D, Grillo MT, Lancellotti P, et al. Right ventricle in pulmonary arterial hypertension: haemodynamics, structural changes, imaging, and proposal of a study protocol aimed to assess remodelling and treatment effects. Eur J Echocardiogr. 2010;11:27–37.
115. Jain A, McNamara PJ. Persistent pulmonary hypertension of the newborn: advances in diagnosis and treatment. Semin Fetal Neonatal Med. 2015;20:262–71.
116. Giesinger RE, More K, Odame J, Jain A, Jankov RP, McNamara PJ. Controversies in the identification and management of acute pulmonary hypertension in preterm neonates. Pediatr Res. 2017;82:901–14.

AUTHOR CONTRIBUTIONS

This document was reviewed and approved by both the Fetus and Newborn Committee at the Canadian Pediatric Society and the Neonatal Hemodynamics Research center. All authors [AJ, REG, SD, YES, RPJ, DEW, SL, SM, MLM, JT, MN, PJM] contributed equally to the intellectual content, writing, and have approved the final version of the manuscript.

COMPETING INTERESTS

The authors declare no competing interests.

ADDITIONAL INFORMATION

Correspondence and requests for materials should be addressed to Patrick J. McNamara.